

Can a machine hear how you feel?

Speech emotion recognition with convolutional neural networks — project and assignment

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Class presentation · assignment due 29/09/2026

What we will cover today

- ▶ **The problem** — emotion in the voice
- ▶ **The data** — five corpora, four languages
- ▶ **From sound to image** — spectrogram and estimated laryngogram
- ▶ **Three CNNs** — scratch, transfer, dilated
- ▶ **Results and a live demo**
- ▶ **The assignment** — groups of 3, due 29/09/2026

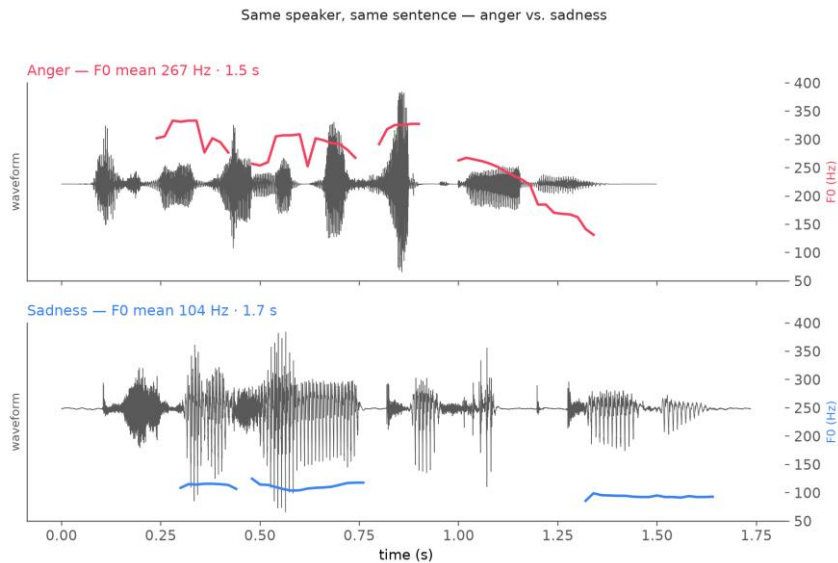
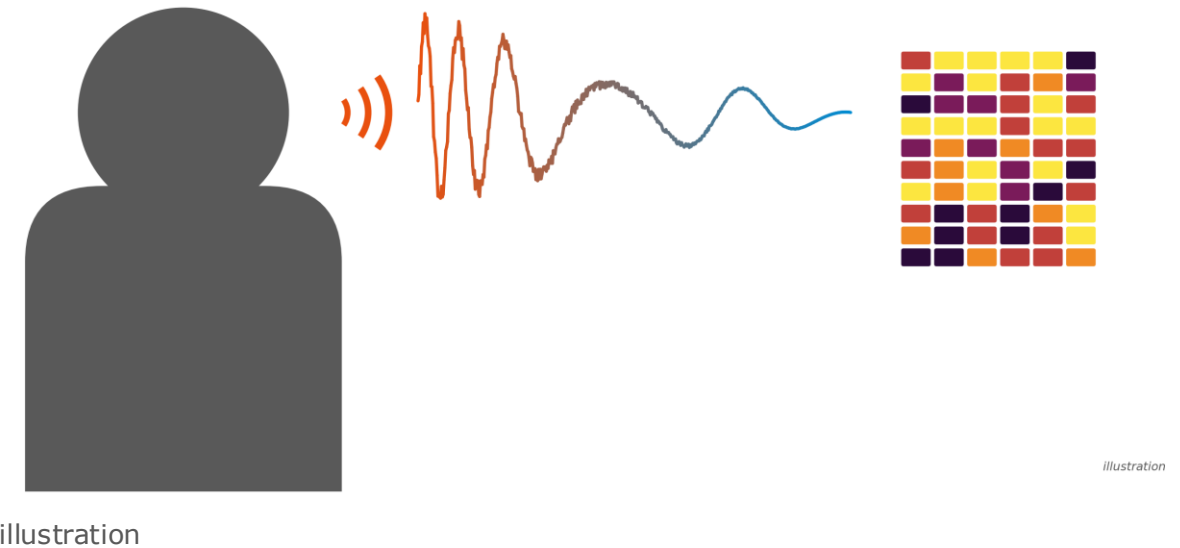
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The problem and the data

What we are trying to recognise, and what we can learn it from.

Emotion lives in how we speak, not only in what we say

- ▶ The same sentence can be angry, sad or joyful — the words do not change, the voice does.
- ▶ **Cues:** pitch and its movement, loudness, speaking rate, voice quality.
- ▶ **Uses:** assistive and social robots, call triage, mental-health monitoring.
- ▶ **Care:** consent, bias across languages and speakers, and never covert use.



The “big six” emotions, plus neutral



Emotion	What the voice tends to do
Anger	Loud, fast, high and widely varying pitch, tense voice
Happiness	Raised pitch, lively contour, bright timbre
Sadness	Quiet, slow, flat and low pitch, breathy
Fear	High pitch, fast, irregular, trembling
Disgust	Slow, lowered pitch, creaky or nasal
Surprise	Sudden pitch jump, short, rising contour
Neutral	The speaker's baseline — the reference for everything else

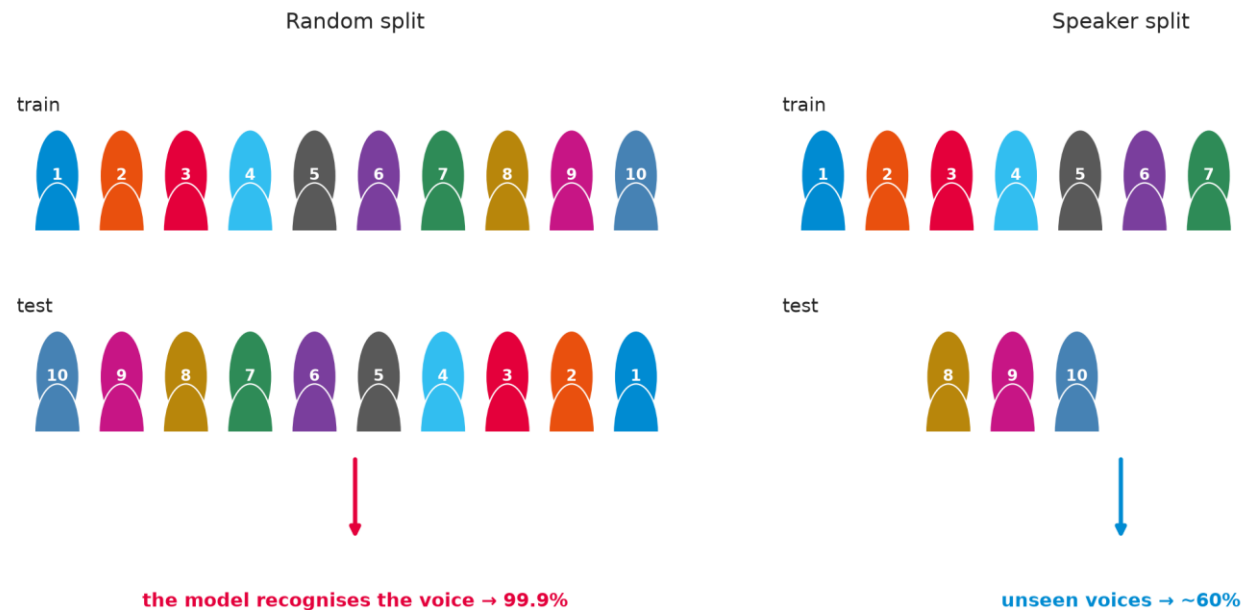
Five corpora, four languages

German		English		Spanish	Portuguese	Multilingual
Corpus	Language	Clips	Speakers	Emotions		
EmoDB	German	535	10	6 + boredom, no surprise		
RAVDESS	English	1,440	24	big six + neutral + calm		
CREMA-D	English	7,442	91	5 + neutral, no surprise		
MESD	Spanish	~1,150	not identified	5 + neutral, no surprise		
emoUERJ	Portuguese	377	8	happiness, anger, sadness, neutral		

Only anger, happiness, sadness and neutral exist in all four languages.

The trap: testing on voices the model already knows

- ▶ Split clips at random and the same actor lands in training and test — the network can recognise the person, not the feeling.
- ▶ That is how a published CNN reaches 99.89 % on a corpus with only two speakers.
- ▶ **Our rule:** no speaker ever appears in two splits.
- ▶ **Few speakers?** Leave-speakers-out cross-validation, so every number is on unseen voices.



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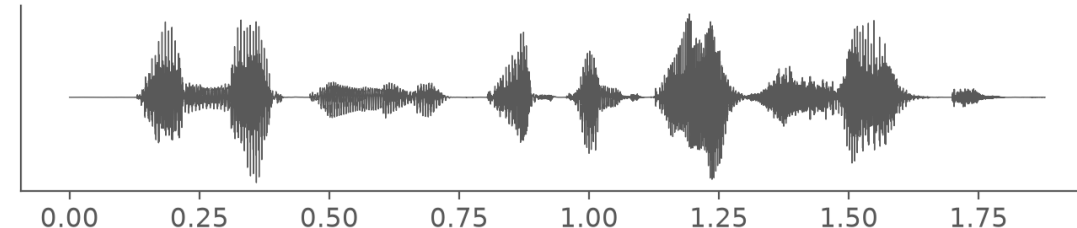
From sound to a prediction

Features, three CNNs, and what they achieve.

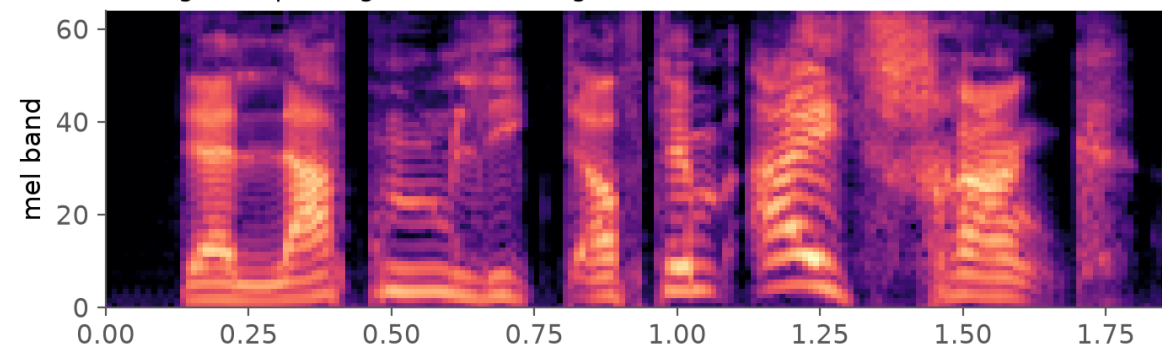
Turning a voice into a picture

- ▶ **Resample** to 16 kHz mono, keep a 3-second window.
- ▶ **Short-time spectrum:** 25 ms frames every 10 ms.
- ▶ **Mel scale:** 64 bands spaced like human hearing, in dB, standardised.
- ▶ **Estimated laryngogram:** inverse filtering strips the vocal tract to reveal the vocal-fold pulses. A real one needs neck electrodes — this is an estimate.

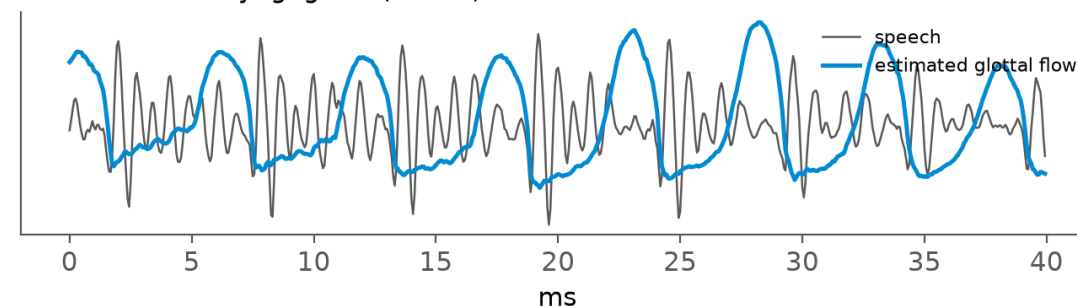
1 · Sound wave



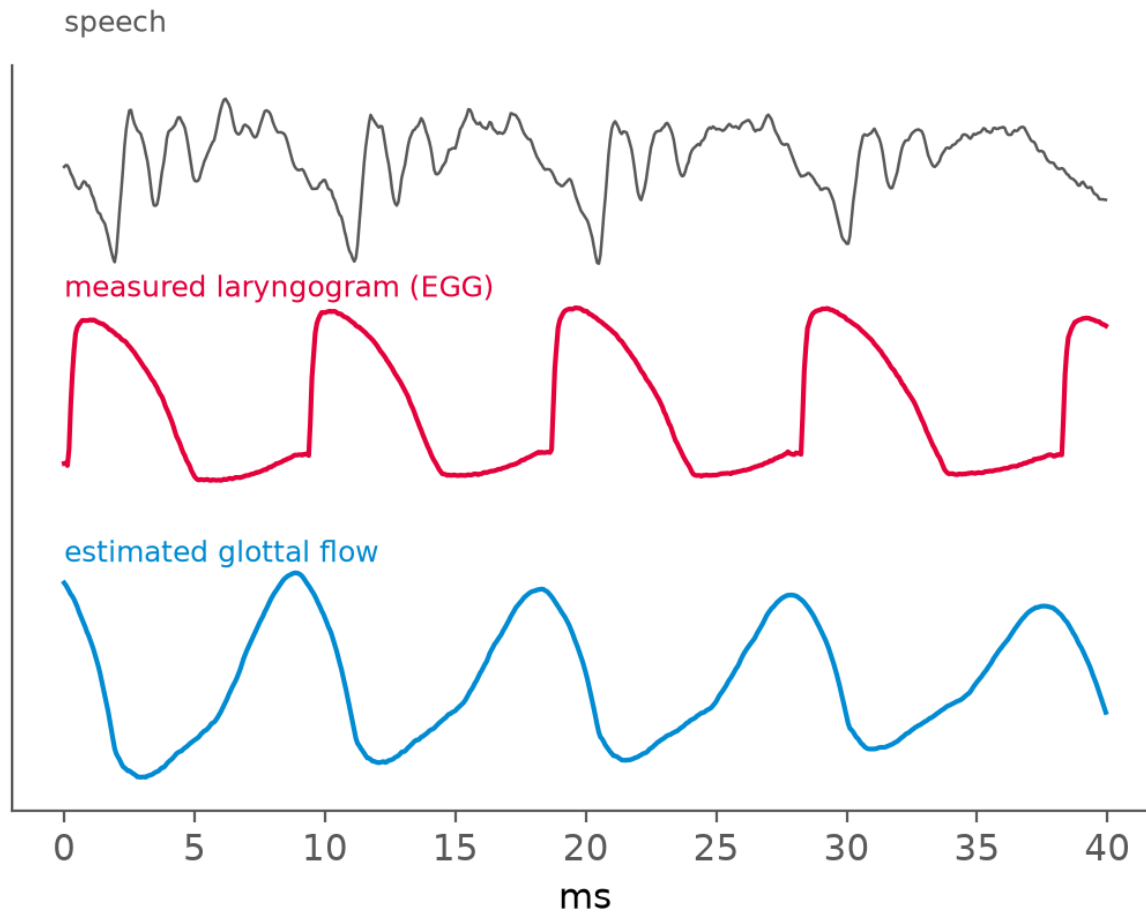
2 · Log-mel spectrogram — the image the CNN reads



3 · Estimated laryngogram (40 ms)



Does the estimate match a real laryngogram?



- ▶ EmoDB includes 792 recordings with a real laryngograph channel — ground truth.
- ▶ Pitch from our estimate vs the measured signal: 1.2 % median error, 2 % gross errors.
- ▶ Waveform agreement (closure events): correlation 0.53.
- ▶ Airflow and vocal-fold contact are different physical signals — timing should match, shape need not.
- ▶ For reference, this correlation measure scores about 0.36 on unrelated smooth signals, so 0.53 indicates genuine but moderate alignment.

EmoDB frame with median agreement — not a hand-picked example.

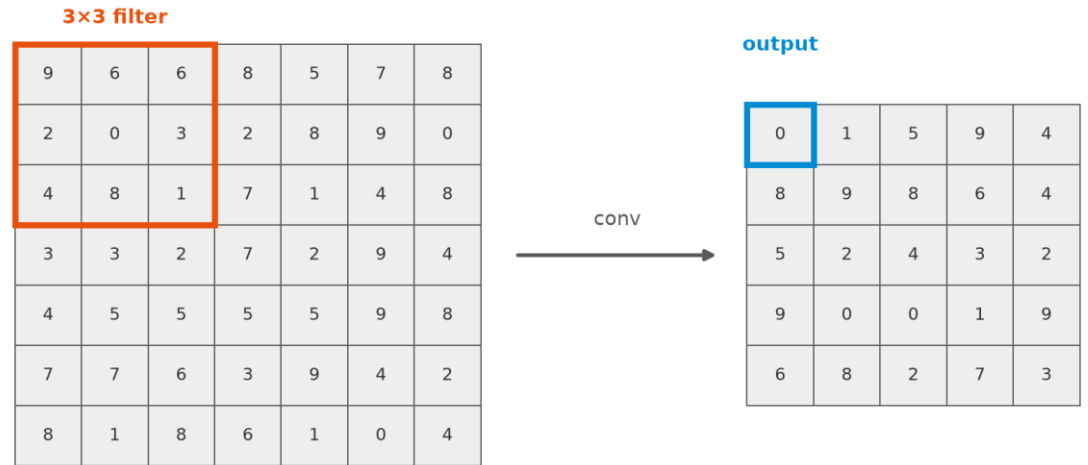
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How a CNN sees

Five ideas that make convolutional networks work on spectrograms.

Convolution: a small filter that slides

- ▶ A 3×3 filter is dotted with every 3×3 patch of the input; the result is one number per position.
- ▶ **Local connectivity:** each output looks at a small neighbourhood, not the whole image.
- ▶ **Parameter sharing:** the same 9 weights are reused at every position — few parameters, translation-equivariant.
- ▶ **Depth:** 32 filters → 32 feature maps; each learns a different pattern.



output = $\sum$ (patch × filter) + bias — the same 9 weights everywhere

Padding, stride and the size of the output

- ▶ Output size = $(W - F + 2P) / S + 1$
- ▶ 64×301 mel image, 3×3 filter, padding 1, stride 1 $\rightarrow 64 \times 301$: 'same' padding keeps the size
- ▶ **Stride 2** halves the map — a cheap way to zoom out

No padding, stride 1: $7 \times 7 \rightarrow 5 \times 5$
 $(W - F + 2P)/S + 1 = (7 - 3 + 2 \cdot 0)/1 + 1 = 5$

8	0	1	2	1	8	8
5	0	0	3	4	6	4
2	1	6	7	0	1	4
3	8	5	4	4	6	5
1	7	7	9	7	2	3
6	6	6	8	2	9	0
0	9	9	2	1	3	0

Same padding, stride 2: $7 \times 7 \rightarrow 4 \times 4$
 $(W - F + 2P)/S + 1 = (7 - 3 + 2 \cdot 1)/2 + 1 = 4$

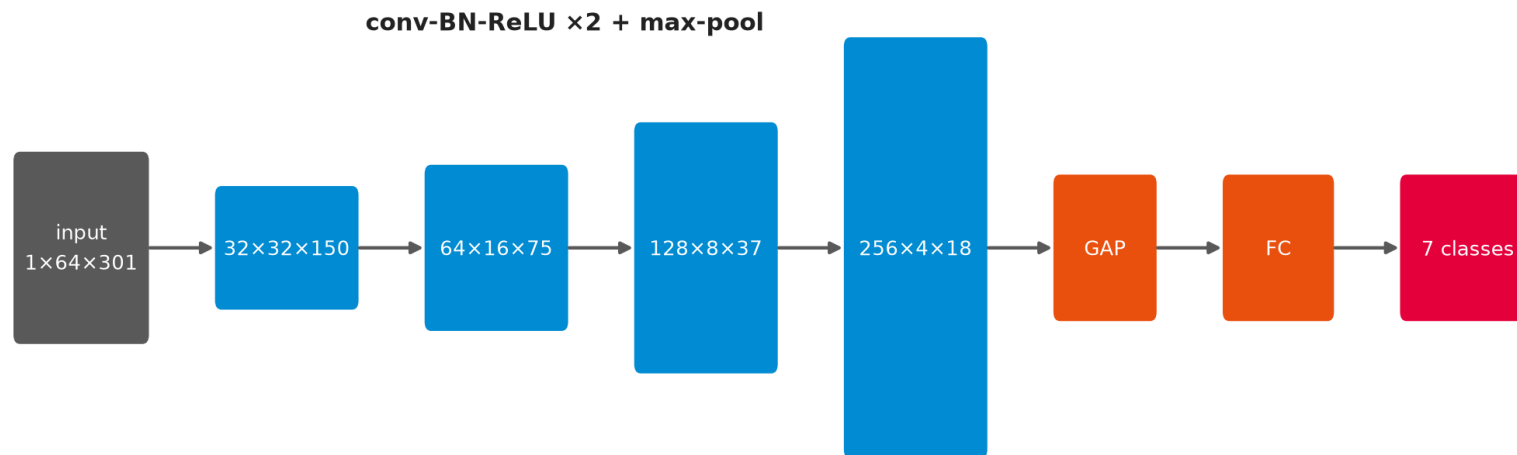
	3	4	6	4	2	1	6
	1	4	3	8	5	4	4
	1	7	7	9	7	2	3
	6	8	2	9	0	0	9
	1	3	0	8	6	5	2
	7	4	0	2	7	5	3
	6	6	5	9	9	2	6

dashed border = zero padding · orange/blue boxes = filter footprint, one stride apart

Redrawn after Dumoulin & Visin 2016 (arXiv:1603.07285)

Pooling and depth: from edges to emotion

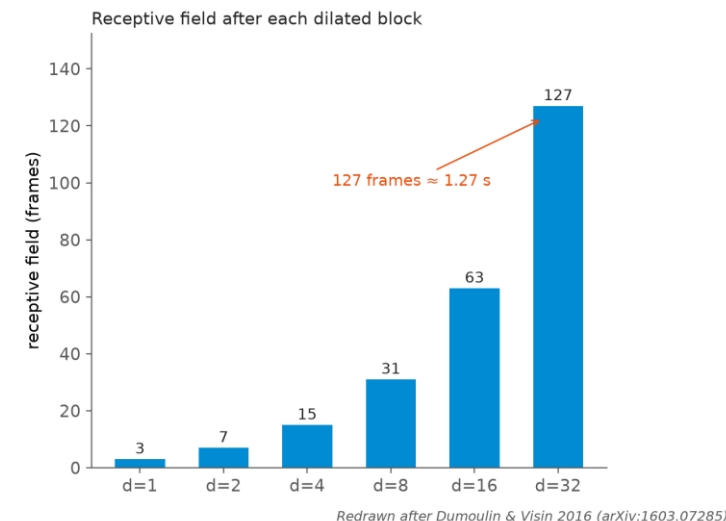
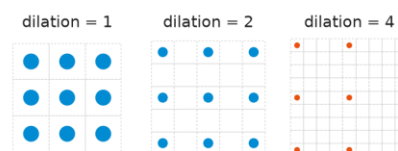
- ▶ **Max-pooling** keeps the strongest response in each 2×2 window: half the size, more robust to small shifts.
- ▶ **Stacking** conv $\rightarrow$ ReLU $\rightarrow$ pool four times builds a hierarchy: edges $\rightarrow$ harmonic stripes $\rightarrow$ syllable-scale patterns.
- ▶ **Global average pooling** turns the last maps into one vector, then a fully connected layer votes for the emotion.
- ▶ This is exactly our Model 1.



Receptive field and dilation: seeing a whole phrase

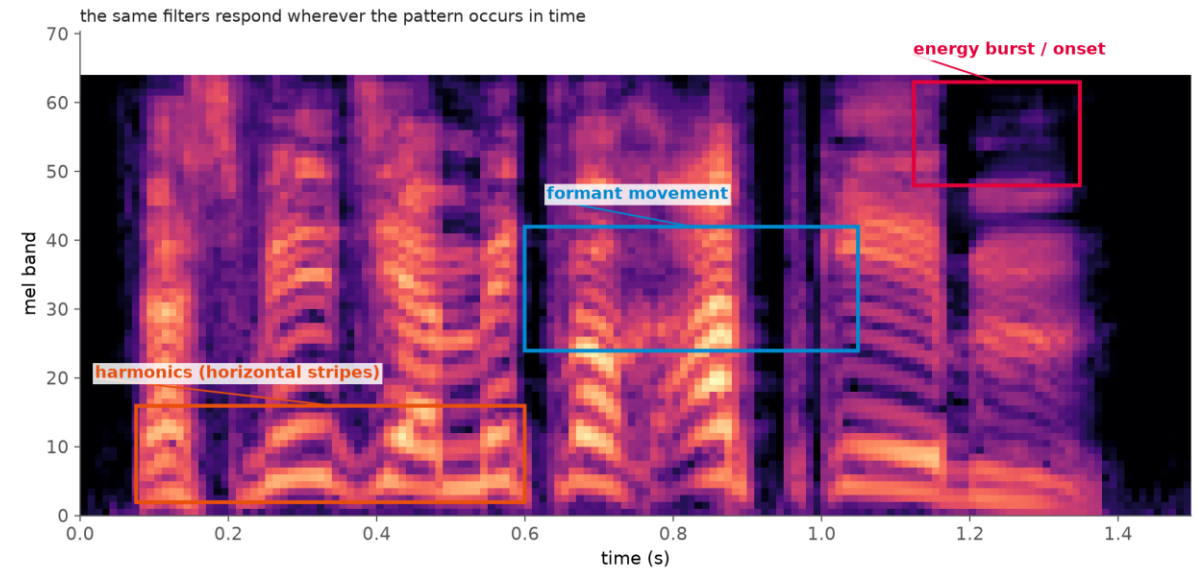
- ▶ **Receptive field:** how much of the input one output value can see; it grows with every layer.
- ▶ **Dilated kernels** skip samples: effective size $k + (k-1)(d-1)$; a 3×3 kernel with dilation 8 spans 17 frames.
- ▶ **Rates 1-2-4-8-16-32** → **127 frames** $\approx$ **1.27 s** of context at 10 ms per frame — a phrase, not a syllable — without pooling away resolution.
- ▶ **Our design:** neither reference paper uses dilation; Model 3 tests whether it helps.

3×3 kernel taps at increasing dilation



Why it works on a spectrogram

- ▶ A spectrogram is an image whose axes are time and frequency.
- ▶ **Harmonics** are horizontal stripes; formant movement bends them; onsets are vertical edges — all local patterns.
- ▶ **Translation in time** is the invariance we want: the same filter fires wherever the pattern occurs.
- ▶ **Frequency is different:** shifting a pattern up changes the pitch, so the network keeps frequency position (we pool it less).



A CNN trained from scratch

- ▶ Four blocks: two 3×3 convolutions + batch-norm + ReLU, then 2×2 max-pooling; $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ channels.
- ▶ Global average pooling $\rightarrow$ dropout $\rightarrow$ one fully connected layer.
- ▶ Small on purpose: 1.2 M parameters, trains on a laptop CPU.
- ▶ Regularised with SpecAugment (masked time and frequency stripes) and random time shifts.

Transfer learning: borrow a bigger network

- ▶ ResNet-18 pretrained on ImageNet: 11 M parameters that already know edges, textures and shapes.
- ▶ The spectrogram is copied into three channels so it looks like an RGB image.
- ▶ **Frozen backbone** — only a new fully connected head is trained, so it fits tiny datasets.
- ▶ Works when spectrogram patterns resemble visual textures; struggles when the domain gap is large.

Dilated CNN: see further without shrinking the picture

- ▶ Dilated kernels skip samples: a 3×3 kernel with dilation 8 spans 17 frames.
- ▶ Rates 1-2-4-8-16-32 along time give a receptive field of about 1.3 s — a whole phrase — with no temporal pooling.
- ▶ Prosody unfolds over hundreds of milliseconds; this keeps frame resolution while seeing that far.
- ▶ **Our own design.** Neither reference paper uses dilation; we test whether it helps.

Where the best CNN gets confused

True	anger	65	10	4	20	1	0	1
	disgust	0	59	9	2	9	2	20
	fear	3	7	72	3	4	7	3
	happiness	39	13	17	23	1	4	3
	sadness	0	5	10	0	77	2	6
	neutral	0	6	8	0	0	59	27
	boredom	0	10	0	0	6	19	65
		anger	disgust	fear	happiness	sadness	neutral	boredom
		Predicted						

- ▶ 39 % of happiness clips are heard as anger.
- ▶ 27 % of neutral clips are heard as boredom.
- ▶ Confusions follow arousal: high-energy emotions are mistaken for each other, and so are low-energy ones.

CNN from scratch · German · row-normalised, %

What we learned



Best CNN on German: CNN from scratch, 60 % unweighted recall on speakers it never heard.



Transfer-learning results are being completed on GPU.

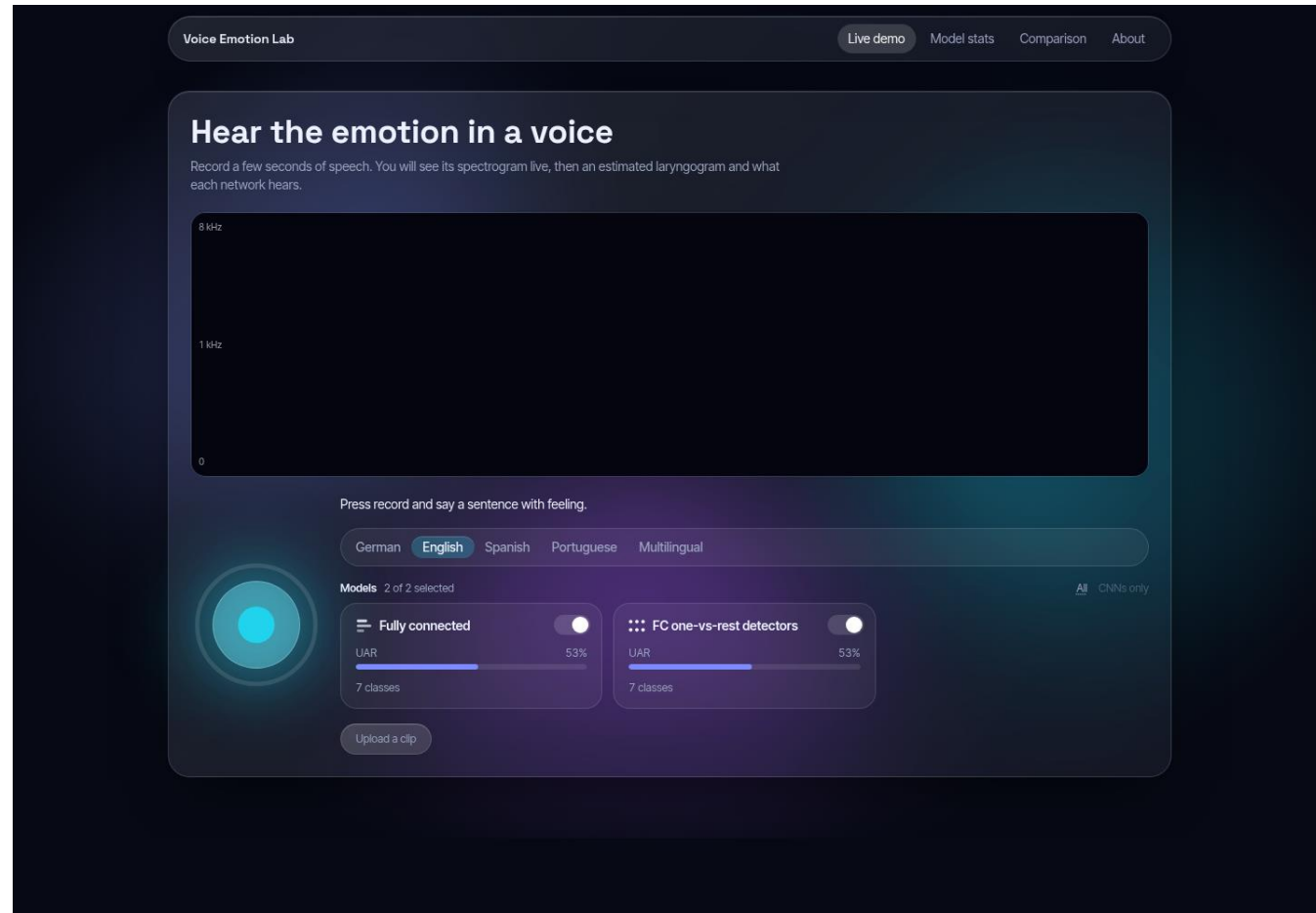


German is the hardest: 60 % with 7 classes. Less data and fewer speakers hurt more than the choice of architecture.

Try it now

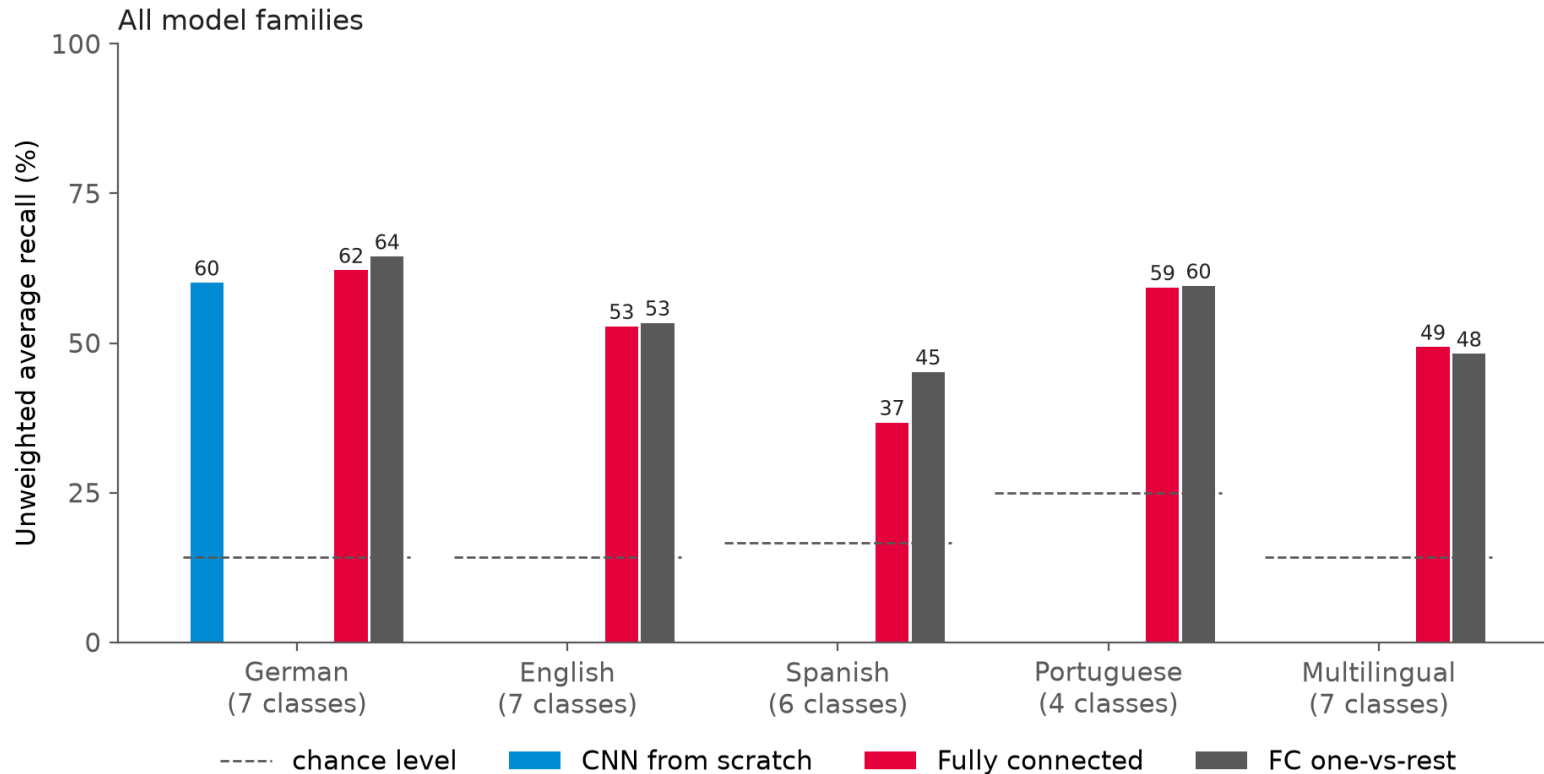


- ▶ **Open** <https://mango-tree-08fd18503.3.azurestaticapps.net>
- ▶ Press record and say a sentence with feeling.
- ▶ Watch the spectrogram scroll live, then see what each network hears.
- ▶ Switch language and compare the models side by side.



the live demo — record, watch the spectrogram, compare models

What else is in the project



- ▶ **Fully connected networks** on 73 handcrafted features: pitch, jitter, shimmer, MFCCs, glottal measures.
- ▶ **One-vs-rest detectors:** a separate small network per emotion.
- ▶ **Next:** transformer speech encoders and diffusion-based augmentation.

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Your assignment

Groups of 3 · CNNs only · due 29/09/2026.

Build your own voice-emotion CNN

- ▶ **Teams:** groups of 3.
- ▶ **Models:** CNNs only — at least one trained from scratch and one using transfer learning.
- ▶ **Data:** at least one public corpus, evaluated on speakers the model never saw.
- ▶ **App:** a deployed web page where anyone can record their voice, see its spectrogram and get a prediction.



Due 29/09/2026 — submit two links: your repository and your live app.

How it will be graded

Criterion	Weight
Working live demo (record → prediction)	30 %
Model quality and honest, speaker-independent evaluation	25 %
Code and repository quality	20 %
Spectrogram / signal visualisation in the app	15 %
README: method, results, who did what	10 %

A one-week plan and where to start

- ▶ **Days 1–2:** download a corpus, make spectrograms, split by speaker.
- ▶ **Days 3–4:** train and evaluate both CNNs.
- ▶ **Day 5:** inference API and a recording page.
- ▶ **Day 6:** deploy.
- ▶ **Day 7:** polish, README, who did what.
- ▶ **EmoDB** database.syntheticspeech.de
- ▶ **RAVDESS** zenodo.org/records/1188976
- ▶ **CREMA-D** github.com/CheyneyComputerScience/CREMA-D
- ▶ **Paper** Mengist et al., Sci. Rep. 2026
- ▶ **Paper** Sareen & Seeja, Procedia CS 2025

Questions?

<https://mango-tree-08fd18503.3.azurestaticapps.net>

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Assignment due 29/09/2026 · repository link + live app link